

The Role of AI in Autonomous Robots

Mini Project Proposal

CodeAlpha Robotics & Automation Internship-Task 4

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1. Introduction

Autonomous robots are machines capable of performing tasks in unstructured or changing environments with little or no direct human control. Where classical industrial robots follow fixed, pre-programmed motion paths, autonomous robots must perceive their surroundings, interpret what they perceive, decide on an action, and execute it, repeating this cycle continuously as conditions change. Artificial intelligence (AI) is the technology that makes this cycle possible at a useful level of reliability. This proposal outlines a mini-project examining how AI techniques, specifically computer vision, machine learning, and path-planning algorithms, are integrated into autonomous robotic systems, along with a proposed small-scale demonstration project and its expected learning outcomes.

2. Problem Statement

Traditional automation performs well in controlled, repetitive environments but struggles when conditions vary: a delivery robot encountering an unexpected obstacle, a warehouse robot facing irregularly shaped inventory, or an agricultural robot navigating uneven terrain. Programming an explicit rule for every possible scenario is impractical. AI addresses this gap by allowing robots to generalize from data and prior experience rather than relying solely on hand-coded rules, enabling them to operate in environments their designers did not fully anticipate.

3. Objectives

- Study the core AI techniques used in autonomous robotics: perception (computer vision, sensorfusion), decision-making (planning and reinforcement learning), and control.
- Analyze how these techniques are combined in a typical autonomous robot's software architecture,commonly described as the sense-think-act loop.
- Propose a small-scale demonstration project that illustrates one complete instance of this loop usingaccessible, low-cost hardware.

- Evaluate the benefits and current limitations of AI-driven autonomy compared to traditional, rule-based robot control.

4. Literature Background

4.1 Perception: Computer Vision and Sensor Fusion

Perception is the stage at which a robot converts raw sensor data, camera images, lidar point clouds, ultrasonic range readings, into a usable representation of its environment. Deep learning-based object detection models, particularly convolutional neural network (CNN) architectures, are now standard for identifying and localizing objects, obstacles, and navigable surfaces in camera images. Because any single sensor has weaknesses (cameras struggle in poor lighting, lidar is expensive and can miss thin or transparent objects), most autonomous robots combine several sensor types through sensor fusion, using probabilistic methods such as Kalman filtering to merge inputs into a single, more reliable estimate of the robot's surroundings.

4.2 Decision-Making: Planning and Reinforcement Learning

Once a robot has a model of its environment, it must decide what to do. Classical path-planning algorithms such as A* and Rapidly exploring Random Trees (RRT) compute a route from the robot's current position to a goal while avoiding known obstacles. These methods work well when the environment is mapped and relatively static. Reinforcement learning (RL) extends decision-making into more dynamic or unfamiliar settings: a robot using RL learns a policy, a mapping from situations to actions, through repeated trial and error, typically in simulation, optimizing for a reward signal such as reaching a goal quickly while avoiding collisions.

4.3 Control

The control layer converts a planned trajectory or high-level action into precise motor commands, joint angles for a robotic arm, or wheel speeds for a mobile robot. While control theory itself predates modern AI, learning-based control (for example, training a neural network to directly map sensor input to motor output) is increasingly used for tasks such as legged locomotion, where the relationship between terrain and stable movement is too complex to model by hand.

4.4 The Sense-Think-Act Loop

These three stages, perception, decision-making, and control, form a continuous loop that runs many times per second on a functioning autonomous robot. AI's role can appear at any stage of this loop, and modern systems increasingly use AI at every stage rather than confining it to perception alone, which was the more common design a decade ago.

5. Proposed Mini Project

Title: Vision-Guided Obstacle-Avoiding Mobile Robot

Concept: A small, wheeled robot platform (for example, a Raspberry Pi or Arduino-based chassis with an ultrasonic sensor and a low-cost camera) that autonomously navigates a room, avoiding obstacles and pausing to identify recognizable objects it passes using a lightweight pre-trained image classification model.

5.1 System Architecture

Layer	Implementation in This Project
Sensing	Ultrasonic distance sensor for close-range obstacle detection; onboard camera for visual input.
Perception	A lightweight pre-trained CNN (such as MobileNet) runs on-device or on a connected laptop, used to classify objects the camera detects.
Decision-making	A simple reactive planning rule set: if an obstacle is detected within a threshold distance, select a turn direction based on which side has more clearance, informed by ultrasonic readings.
Control	Motor driver commands translating the chosen direction into differential wheel speeds.
Feedback loop	The system re-evaluates its surroundings continuously (several times per second), repeating the sense-think-act cycle as it moves.

5.2 Methodology

1. Assemble the robot chassis and wire the ultrasonic sensor, camera module, and motor driver to the onboard microcontroller/single-board computer.
2. Implement the reactive obstacle-avoidance logic and test it in isolation using only ultrasonic input, confirming the robot reliably turns away from obstacles.
3. Integrate the pre-trained image classification model, and verify it correctly labels a small test set of common objects (chair, box, bottle, person) under the robot's onboard camera and lighting conditions.
4. Combine both subsystems so the robot navigates continuously while logging any objects it identifies along its path.

5. Run trials in a controlled indoor space and record navigation success rate (percentage of obstacles successfully avoided) and object-classification accuracy.

5.3 Expected Outcomes

The completed prototype should demonstrate a robot that can traverse a room with scattered obstacles without collision in the large majority of trial runs, while correctly identifying common objects a majority of the time under normal indoor lighting. Beyond the physical prototype, the project is expected to produce a clear, working example of how the sense-think-act loop described in Section 4.4 is implemented with genuinely low-cost, accessible hardware, making it a suitable reference project for further coursework or portfolio use.

6. Benefits and Limitations of AI-Driven Autonomy

6.1 Benefits

- Generalization: AI-based perception can recognize objects and situations it was not explicitly programmed to expect, unlike fixed rule-based systems.
- Adaptability: reinforcement-learned or continuously updated models can improve performance overtime as more data becomes available.
- Reduced engineering overhead for complex environments, since developers do not need to anticipate and hand-code every possible scenario.

6.2 Limitations

- AI models can fail unpredictably on inputs that differ from their training data ("edge cases"), which is a particular concern in safety-critical robotics applications.
- Training data-hungry models requires significant data collection and, for physical robots, either extensive real-world trials or high-fidelity simulation.
- Computational cost: running deep learning models in real time on embedded hardware requires careful optimization, and more capable models often exceed the processing budget of small, low-power robot platforms.
- Explainability: it can be difficult to determine exactly why a learned model made a particular decision, which complicates debugging and safety certification.

7. Conclusion

Artificial intelligence has moved from a peripheral enhancement to a foundational component of autonomous robotics, underpinning perception, decision-making, and increasingly, control. The proposed vision-guided obstacle-avoiding robot project offers a compact, achievable way to implement and observe this sense-think-act loop directly, using accessible hardware and a lightweight pre-trained

model rather than requiring large-scale infrastructure. The exercise is intended to build practical understanding of how the AI techniques discussed in this proposal come together in a working system, while also surfacing the real limitations, data dependency, computational cost, and unpredictability on edge cases, that any engineer working in this field needs to account for.

References

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